

Fly GACA

Master Plan

Building Saudi aviation's regulatory library and AI flight instructor — product and business, from scratch.

"The whole of Saudi aviation, in one cockpit."

Prepared for Captain Adel Al-Subaie

Version 1.0 · May 2026

Build mode Solo founder · no fixed deadline · phased delivery

Primary domain flygaca.com · server 72.62.20.20

Companion files: this document is the human-readable master plan. Two AI briefings package the same plan as ready-to-use context — one XML-tagged for Claude, one Markdown (AGENTS.md) for Google Antigravity.

Contents

OVERVIEW

1 · Executive summary

Fly GACA is an educational digital aviation platform and open regulatory library for Saudi Arabian civil aviation — built for pilots, operators, instructors, and cadets in the Kingdom. This plan covers building it from scratch, product and business together, as a solo effort with no fixed deadline.

The strategy is a single funnel. A free, fast, genuinely useful GACAR library earns reach and trust. Captain Adel — an AI flight instructor that answers regulatory questions with cited sources — is the paid premium layer. Flight schools and operators buying seats for their cadets are the revenue engine.

The build is sequenced into seven phases, 0 through 6. Each phase ships something usable on its own, and risk rises gradually: the public library launches first with no login and no AI; personal data is not touched until compliance is ready for it; monetization arrives only once the assistant is proven good.

The recommended infrastructure is managed-first — Firebase, with the database in the me-central2 (Dammam) region for data sovereignty — with the existing VPS at 72.62.20.20 repurposed as the document-processing pipeline rather than the production front door.

Phase 0 gate — Three questions must be answered before heavy building begins: whether the GACAR documents can be legally rehosted, the trademark exposure of the “GACA” name, and which domain is canonical.

DIRECTION

2 · Strategic frame

The wedge

Saudi pilots juggle scattered PDFs, paper logbooks, calculators, and a dozen reference sites just to study and stay current. Fly GACA collapses that into one cockpit. The free GACAR library is the wedge — it is the lowest-risk, highest-trust thing to build, and it is useful on its own. Trust earned there converts into paid use of Captain Adel and the pilot tools; flight-school and operator accounts are where the revenue concentrates.

Captain Adel — one service, two front doors

Captain Adel is built **once**, as a single assistant service. It is served inside flygaca.com behind the paywall, and reachable through captadel.com as a marketing front door. One brain, two doors — not two codebases. Whether Captain Adel is ultimately a sub-brand of Fly GACA or a standalone product is deliberately deferred until the assistant works; the build does not depend on that decision.

Scope discipline

The project brief describes a broad feature set. For a solo build, the version that goes live first is the library and search — nothing else. The assistant, accounts, tools, payments, and native apps are all post-launch, sequenced in the phases below. Resisting scope creep is the single most important habit for shipping this solo.

RESOLVE FIRST

3 · Three risks to clear before building

These shape Phase 0. They are flagged from an engineering and product view — **not legal advice**; a Saudi IP lawyer should confirm items 1 and 2.

3.1 Redistribution rights for the documents

This is the genuine launch-blocker. The platform’s core is hosting GACA-published documents — the GACAR parts, the AIPs, and advisory circulars. Official government documents are, in many jurisdictions and under Saudi copyright practice, outside copyright protection — but confirm the specific Saudi position, or obtain GACA’s permission, before rehosting the corpus.

Interim safe posture: source every document, show its official version and date, and link to GACA’s own copy. If clearance is not yet in hand, the library can launch as a deep-linking index and rehost once confirmed.

3.2 The name leans on “GACA”

“Fly GACA” borrows the identity of the actual government authority. Even with disclaimers, this carries trademark and passing-off exposure and can confuse users into assuming official status. Run a trademark check, and carry an unmissable “not affiliated with the General Authority of Civil Aviation” disclaimer on every page and in the assistant’s answers.

3.3 Domain canonicalization

flygaca.com is owned and live on 72.62.20.20; the brand sheet repeatedly shows flygaca.sa. Choose one canonical domain now — favicon copy, Open Graph cards, email addresses, and business cards all depend on it. The plan assumes flygaca.com unless changed.

KNOWLEDGE BASE

4 · Captain Adel corpus policy

What Captain Adel is allowed to learn from is a legal and safety decision, not just a technical one. The corpus splits three ways.

Tier	Policy
Host — safe core	GACAR parts, Saudi AIPs, and GACA advisory circulars / guidance. All GACA-published; one legal clearance covers them. This is the foundation.
Host — original	Fly GACA’s own training and exam material, authored from the Captain Adel Curriculum Map. Legal because it is yours, and a real differentiator. Scraped third-party question banks are not permitted.
Do not host — cite & refer	ICAO Annexes / SARPs (ICAO copyright; sold commercially — and their substance is already reflected in GACAR). Aircraft manuals, POH / AFM (manufacturer copyright, and aircraft-specific data is a safety hazard if quoted generically). Captain Adel cites and points to the official source; it never reproduces or substitutes for these.

AIP freshness rule — AIPs change every 28-day AIRAC cycle. Every AIP-sourced answer must carry an effective date and the line “not for operational use — verify the current AIP and NOTAMs.”

INFRASTRUCTURE

5 · Hosting & infrastructure decision

Recommendation: managed-first. Run the platform on Firebase, with the existing VPS repurposed — not as the production front door.

A solo founder’s scarcest resource is time. Self-hosting authentication, a database, and security for a platform that will hold pilots’ personal data (PDPL) is a large, permanent, security-sensitive burden. Firebase provides App Check, Authentication, security rules, and a Firestore database in the me-central2 (Dammam) region out of the box — exactly the “harden by default” and data-sovereignty posture the project requires.

Role of the VPS (72.62.20.20). Not wasted. It is the right home for the Python document-ingestion and RAG-indexing pipeline — a batch job that does not need 24/7 uptime guarantees — and for running the assistant eval harness and staging builds. It is a compute box, not the live site.

To verify. Which region the VPS sits in, and the data-residency of the AI inference (Gemini / Vertex). The public regulatory corpus is low-risk; the question only sharpens if personal data is ever sent to the model. When the front door moves to Firebase Hosting, flygaca.com’s DNS migrates off the single A-record to 72.62.20.20.

6 · The build plan — seven phases

Each phase ends with something usable. Phases are sequential by dependency, not by date — there is no deadline; the order is what matters.

Phase	Focus	What it ships
0	Foundations	Cleared blockers, registered entity, empty repo + Firebase project
1	The library goes live	flygaca.com — browsable, searchable GACAR library; no login, no AI
2	Captain Adel answers	The RAG assistant with cited answers; free / beta
3	Pilot accounts & tools	Accounts, digital logbook, currency tracking, core calculators
4	Arabic, offline & polish	Full bilingual RTL PWA, offline library, performance
5	Money & flight schools	Subscriptions, e-invoicing, flight-school / operator accounts
6	Reach	Native app wrapper, public API

Phase 0 — Foundations

Goal. Clear the blockers and stand up the skeleton. No public output.

Key work.

- Resolve redistribution rights for GACAR, AIPs, and advisory circulars (the launch-blocker).
- Pick the canonical domain (flygaca.com) and lock the legal / brand name.
- Register the legal entity — a Saudi establishment or LLC via the Saudi Business Center; consider Monshaat SME registration and the NTDP program. Open a business bank account.
- Create the GitHub repository (github.com/ay2m/flygaca) with the agreed folder structure.
- Create the Firebase project in me-central2 (Dammam). Set up the VPS as the pipeline / staging box.

Done when — The copyright question has an answer, the entity is registered, and the empty repo and Firebase project exist.

Phase 1 — The library goes live

Goal. A fast, trustworthy public GACAR library at flygaca.com. No login, no AI — a pure trust-builder.

Key work.

- Build the static PWA shell on the Falcon brand (Falcon Night dark theme, the teal / sage palette, Inter Tight).
- Curate and text-extract the GACAR corpus; build the library index and a clean document reader.
- Build client-side search over a prebuilt index (the corpus is small enough for this).
- Apply brand assets: favicon, app icon, Open Graph card, the approved lockups.
- Ship the legal pages: educational disclaimer, “not affiliated with GACA” notice, terms, attributions.
- Deploy to Firebase Hosting; point flygaca.com DNS.

Done when — A visitor can search “night currency,” open Part 91, read it on a phone — and every page makes the unofficial, educational status obvious.

Phase 2 — Captain Adel answers

Goal. Plain-language Q&A grounded in the corpus, with cited passages and source links.

Key work.

- Build the ingestion pipeline on the VPS (Python 3.11 — chunk and embed the corpus).
- Choose and stand up a vector store; build the RAG Cloud Function (retrieve → prompt → Gemini → answer + citations).
- Build the chat UI (English and Arabic) at chat.html, with captadel.com as a front door.
- Write Captain Adel’s system prompt: conservative, always cites, always defers to GACA, refuses to substitute for a POH / AFM.
- Build the eval harness with question sets so retrieval quality is measured, not guessed.
- Enforce App Check and per-user rate limits so the AI endpoint cannot be abused or run up costs.

Done when — The eval set passes its quality bar, every answer carries citations, and no answer ever asserts official status. Launch free / beta — paywall comes in Phase 5, once quality is proven.

Phase 3 — Pilot accounts & tools

Goal. Personalised pilot features — the first time the platform holds personal data.

Gated: the PDPL compliance program and DPIA must be signed off before this phase ships.

Key work.

- Firebase Authentication; a Firestore data model with strict per-user security rules; App Check enforced.
- Digital logbook: log flights, track currency per GACAR (VFR / IFR / night / landings).
- Core calculators — weight & balance first, then one or two more high-value ones.

- Profile and settings; data export and full account deletion (PDPL data-subject rights).

Done when — A pilot can register, log a flight, see currency status, export a report, and fully delete their account and data.

Phase 4 — Arabic, offline & polish

Goal. A complete, resilient bilingual progressive web app.

Key work.

- Full Arabic RTL parity across the UI (the brand already provides RTL lockups).
- Offline-aware service worker so the library is readable without signal — genuinely valuable for pilots.
- Accessibility pass and Web Vitals; Playwright end-to-end test coverage; the error beacon and observability.

Done when — The whole UI works in Arabic RTL, the library works offline, and performance scores are green.

Phase 5 — Money & flight schools

Goal. Turn on revenue — Captain Adel becomes the paid product.

Key work.

- Define the pricing model and build pricing.html; set the free-vs-paid split with the assistant behind the paywall.
- Integrate a Saudi payment gateway (e.g. Moyasar or Tap, SAR-native) wired to ZATCA Fatoora e-invoicing.
- Build subscription and billing logic; handle VAT.
- Flight-school and operator accounts (schools.html, admin view): seat management for cadets, instructor oversight.

Done when — A pilot can subscribe and receive a compliant e-invoice, and a flight school can buy and manage cadet seats.

Phase 6 — Reach

Goal. Extend distribution beyond the web app.

Key work.

- Capacitor native wrapper for the iOS and Android app stores.
- A documented public API for third-party integrations.

Done when — The app is published in both stores and the public API has documented, versioned endpoints.

COMPANY

7 · Business & company track

This runs alongside the build. Much of it is finalizing drafts that already exist in the project knowledge base — the strategic plan and OKRs, the PDPL program and DPIA, the ZATCA pack, the Monshaat SME kit, the NTDP application, and the founders / ESOP / IP agreements — rather than creating them from nothing.

- **Entity & registration (with Phase 0).** Register via the Saudi Business Center; complete Monshaat SME registration; pursue NTDP support; open a business bank account.
- **Legal documents (with Phase 1).** Finalize the terms, EULA, disclaimers, and privacy notice once the document-redistribution question is settled.
- **PDPL compliance (before Phase 3).** Sign off the PDPL compliance program and the DPIA. This is the gate for handling personal data.
- **Tax & e-invoicing (with Phase 5).** ZATCA: VAT registration (mandatory above SAR 375k revenue, voluntary from SAR 187.5k) and Fatoora e-invoicing integration.
- **Ongoing.** Bookkeeping and the monthly close; keep a decision log so the rationale behind big choices survives.

GROWTH

8 · Go-to-market

Saudi aviation is a small, reachable audience — which is an advantage. The brand system already co-brands “Captain Adel · on duty” as the face of the product, and that is the moat: a real, credentialed instructor (Chief Instructor, ATPL, CFII) standing behind every answer.

The play. Publish regulatory explainers from Captain Adel’s LinkedIn (sa.linkedin.com/in/captadel) to build authority and funnel readers to the free library. Seed directly with flight schools and pilot communities. Capture a waitlist from the day the library launches.

The funnel. The free library earns reach and trust; Captain Adel and the pilot tools convert individuals; flight-school and operator seats are where the revenue concentrates.

REFERENCE

9 · Tech stack & repository

Layer	Choice
Frontend	Vanilla JavaScript (ES2022), HTML5, CSS3, PWA — no framework.
Backend	Firebase Cloud Functions (Node.js 20).
Database	Cloud Firestore, me-central2 (Dammam) region.
Auth & security	Firebase Authentication, App Check, strict security rules.

Layer	Choice
AI assistant	Captain Adel — Gemini, with a custom system prompt and a RAG pipeline.
Document pipeline	Python 3.11 for chunking and indexing; Node.js builders.
Hosting & CI/CD	Firebase Hosting + GitHub Actions.
Observability	Web Vitals and a custom error beacon.
Testing	Playwright end-to-end; a custom eval harness for the assistant.

Repository (github.com/ay2m/flygaca): assets/ · assistant/ · build_finance/ · evals/ · functions/ (with functions/rag/) · library/PDFs/ · legal/ · office/ · tests/ · .github/ — plus the top-level HTML pages (flygaca, dashboard, library, lessons, exam, chat, pricing, schools, about, admin, settings).

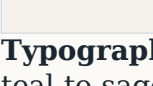
START HERE

10 · Immediate next actions

1. Confirm flygaca.com as the canonical domain.
2. Resolve the GACAR / AIP / advisory redistribution question — the one launch-blocker.
3. Decide the legal entity and lock the brand / company name.
4. Confirm the hosting recommendation (managed-first, Firebase + me-central2, VPS as the pipeline box).
5. Create the GitHub repository, the Firebase project, and the VPS pipeline box.
6. Curate the GACAR corpus — then begin the Phase 1 build.

A · Brand quick-reference

Carried from the Fly GACA Brand Identity Sheet (Mark v2, Falcon-over-Kingdom) so whichever tool builds the interface has the essentials.

	Colour	Hex	Role
	Falcon Night	#0A0E12	Primary canvas — the deck the falcon rests on.
	Falcon Teal	#2D6E8A	Primary brand colour. Buttons, links, focus.
	Falcon Sage	#8FC9A8	Secondary accent. Success states.
	Falcon Gold	#C8A04A	Heritage accent — eyebrows and version stamps only, sparingly.
	Teal Bright	#4A9CB8	Hover, links on dark, focus rings.
	Ivory	#F5F2ED	Reading surface — documents and library.

Typography. Inter Tight throughout; the wordmark sets “Fly” solid and “GACA” in a teal-to-sage gradient.

The mark. Two falcon wings inside the silhouette of the Kingdom, with the letter F in the negative space. Never stretch, skew, rotate, recolour, or outline it; keep at least 4.5:1 contrast against any background; minimum size 16 px; preferred surface Falcon Night.